

Challenge Write-Up Template

Challenge Information

Student Name	Pascua, Diana KC S.
Section / Class	BSIT-3G
Challenge Name	Piece by Piece
Category	General Skills
Points / Difficulty	Easy
Date Solved	8/17/26
Flag Format Given	picoCTF{...}

1. Challenge Description

This challenge gave me SSH access to a remote server where, according to the prompt, my home directory would contain "multiple file parts" that needed to be combined and extracted to reveal the flag. In other words, I wasn't handed one broken or hidden file — I was handed several fragments of a file that had deliberately been split apart, and the job was to figure out how they went back together and what format the reassembled file was actually in.

2. Initial Analysis / Reconnaissance

After logging in with `ssh ctf-player@dolphin-cove.picoctf.net -p 55048`, the first thing I did was run `ls -la` to see exactly what I had. Instead of one file, there was an `instructions.txt` and five oddly-named files: `part_aa`, `part_ab`, `part_ac`, `part_ad`, and `part_ae`. That naming convention — a common prefix with sequential two-letter suffixes — is the classic output of the Unix `split` command, which chops a large file into smaller, evenly-sized chunks with names like `xaa`, `xab`, `xac` and so on. That told me right away this was a file-reconstruction challenge rather than anything cryptographic or web-based: my job was to concatenate the pieces back into their original order and then figure out what the resulting file actually was.

3. Tools and Commands Used

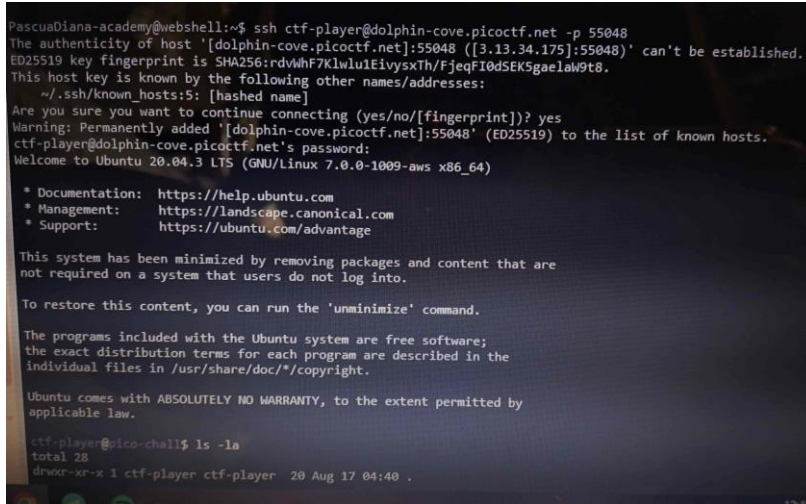
Tool / Command	What it was used for	Result / Output
<code>ssh ctf-player@dolphin-cove.picoctf.net -p 55048</code>	Connect to the challenge server with the credentials picoCTF provided.	Logged into an Ubuntu 20.04.3 LTS box as ctf-player, dropped into a home directory I hadn't looked at yet.
<code>ls -la</code>	See what I actually had to work with before doing anything else.	An instructions.txt file plus five similarly-sized files named part_aa, part_ab, part_ac, part_ad, and part_ae — a strong hint that a single file had been split into pieces.
<code>cat part_aa part_ab part_ac part_ad part_ae > combined</code>	Reassemble the parts back into one file, in order, by concatenating them.	First attempt failed with "cat: part-ad: No such file or directory" — I'd typed a hyphen instead of an underscore. Reran it with the correct part_ad and it produced a file named combined with no errors.
File combined	Confirm what type of file the reassembled data actually was before trying to open it.	combined: Zip archive data, at least v1.0 to extract — so the split pieces added back up to a valid ZIP file.
<code>unzip -P supersecret combined -d extracted</code>	Extract the ZIP, supplying the password mentioned in instructions.txt so unzip wouldn't just prompt and hang.	Archive: combined — extracting: extracted/flag.txt
<code>cat extracted/flag.txt</code>	Print the extracted file's contents.	picoCTF{...}

4. Solution Walkthrough

Step 1

Look at what I was given

ls -la showed the instructions.txt file alongside five part_* files, all roughly the same size except the last one (part_ae, which was smaller — consistent with it being the final, leftover chunk of a split file rather than a full-size piece).



```
PascuaDiana-academy@webshell:~$ ssh ctf-player@dolphin-cove.picoctf.net -p 55048
The authenticity of host '[dolphin-cove.picoctf.net]:55048 ([3.13.34.175]:55048)' can't be established.
ED25519 key fingerprint is SHA256:rdVWhF7K1wlu1EivysxTh/FJeqFI0dSEK5gaelaw9t8.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[dolphin-cove.picoctf.net]:55048' (ED25519) to the list of known hosts.
ctf-player@dolphin-cove.picoctf.net's password:
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 7.0.0-1009-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ctf-player@pico-chall$ ls -la
total 28
drwxr-xr-x 1 ctf-player ctf-player 20 Aug 17 04:40 .
```

Step 2

Reassemble the parts in order

I used `cat part_aa part_ab part_ac part_ad part_ae > combined` to concatenate the five pieces back into a single file called `combined`, listing them in alphabetical/sequential order since that's how `split` names its output and the only order that would reconstruct the original bytes correctly. My first attempt actually failed — I fat-fingered `part-ad` with a hyphen instead of an underscore and got "No such file or directory" — a good reminder to read error messages carefully instead of assuming the command itself is wrong. Fixing the typo and re-running the command completed cleanly.

Step 3

Identify what I'd actually rebuilt

Rather than guess at an extension, I ran `file combined`, which inspects a file's actual contents (magic bytes) instead of trusting a name. It reported "Zip archive data, at least v1.0 to extract" — confirmation that the five parts really did add up to a complete, valid ZIP archive, and that my reassembly step hadn't introduced any corruption.

Step 4

Extract the archive

instructions.txt (visible in the directory listing) supplied the archive's password, so I ran `unzip -P supersecret combined -d extracted` to extract combined into a new `extracted/` folder using that password non-interactively. The output confirmed it pulled out a single file: `extracting: extracted/flag.txt`.

Step 5

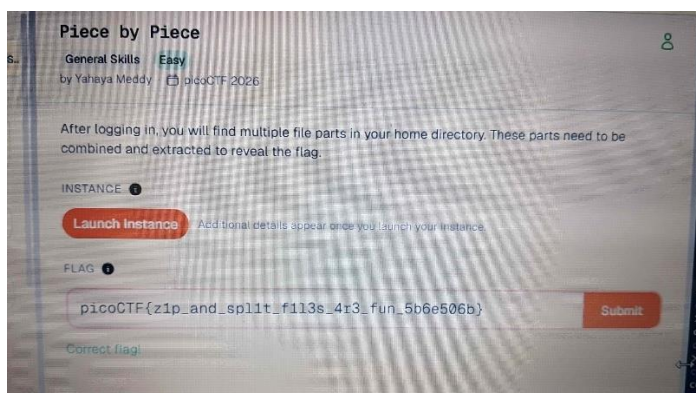
Read the flag

`cat extracted/flag.txt` printed `picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_5b6e506b}` straight to the terminal, which I submitted on the challenge page for a confirmed "Correct flag!"

```
the programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ctf-player@pico-chall$ ls -la
total 28
drwxr-xr-x 1 ctf-player ctf-player 20 Aug 17 04:40 .
drwxr-xr-x 1 root      root      24 Feb  4  2026 ..
drwx----- 2 ctf-player ctf-player 34 Aug 17 04:40 .cache
-rw-r--r-- 1 root      root      67 Feb  4  2026 .profile
-rw-r--r-- 1 ctf-player ctf-player 282 Feb  4  2026 instructions.txt
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4  2026 part_aa
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4  2026 part_ab
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4  2026 part_ac
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4  2026 part_ad
-rw-r--r-- 1 ctf-player ctf-player 35 Feb  4  2026 part_ae
ctf-player@pico-chall$ cat part_aa part_ab part_ac part_ad part_ae > combined
cat: part-ad: No such file or directory
ctf-player@pico-chall$ cat part_aa part_ab part_ac part_ad part_ae > combined
ctf-player@pico-chall$ file combined
combined: Zip archive data, at least v1.0 to extract
ctf-player@pico-chall$ unzip -P supersecret combined -d extracted
Archive:  combined
  extracting: extracted/flag.txt
ctf-player@pico-chall$ cat extracted/flag.txt
picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_5b6e506b}ctf-player@pico-chall$ Connection to dolphin-cove.picoctf.net closed by r
emote host.
Connection to dolphin-cove.picoctf.net closed.
PascuaDiana-academy@webshell:~$
```



5. Flag

```
picoCTF{z1p_and_sp11t_f1l3s_4r3_fun_5b6e506b}
```

6. Key Concept / What I Learned

The core skill here is basic file forensics with everyday Unix tools: recognizing the split-file naming pattern (xaa, xab, ...), knowing that cat can losslessly reassemble binary data as easily as text as long as the pieces stay in order, and using file to identify a file's real type from its magic bytes instead of trusting a filename or lack of extension. It's a good reminder that CTF challenges don't always hide data behind exotic tools — sometimes the whole puzzle is just putting ordinary pieces back together in the right order.

7. References
